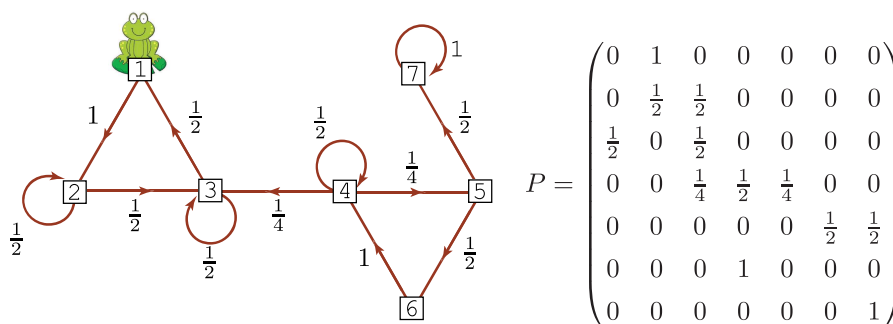


1 Definitions, basic properties, the transition matrix

Markov chains were introduced in 1906 by Andrei Andreyevich Markov (1856–1922) and were named in his honor.

1.1 An example and some interesting questions

Example 1.1. A frog hops about on 7 lily pads. The numbers next to arrows show the probabilities with which, at the next jump, he jumps to a neighbouring lily pad (and when out-going probabilities sum to less than 1 he stays where he is with the remaining probability).



There are 7 ‘states’ (lily pads). In matrix P the element p_{57} ($= 1/2$) is the probability that, when starting in state 5, the next jump takes the frog to state 7. We would like to know where do we go, how long does it take to get there, and what happens in the long run? Specifically:

- Starting in state 1, what is the probability that we are still in state 1 after 3 steps? ($p_{11}^{(3)} = 1/4$) after 5 steps? ($p_{11}^{(5)} = 3/16$) or after 1000 steps? ($\approx 1/5$ as $\lim_{n \rightarrow \infty} p_{11}^{(n)} = 1/5$)
- Starting in state 4, what is the probability that we ever reach state 7? ($1/3$)
- Starting in state 4, how long on average does it take to reach either 3 or 7? ($11/3$)
- Starting in state 2, what is the long-run proportion of time spent in state 3? ($2/5$)

Markov chains models/methods are useful in answering questions such as: How long does it take to shuffle deck of cards? How likely is a queue to overflow its buffer? How long does it take for a knight making random moves on a chessboard to return to his initial square (answer 168, if starting in a corner, 42 if starting near the centre). What do the hyperlinks between web pages say about their relative popularities?

Is the probability distribution of the initial random variable X_0

1.2 Definitions

Let I be a countable set, $\{i, j, k, \dots\}$. Each $i \in I$ is called a **state** and I is called the **state-space**.

We work in a **probability space** $(\Omega, \mathcal{F}, P)$. Here Ω is a set of outcomes, $\mathcal{F}$ is a set of subsets of Ω , and for $A \in \mathcal{F}$, $P(A)$ is the probability of A (see Appendix A).

The object of our study is a sequence of random variables $X_0, X_1, \dots$ (taking values in I) whose joint distribution is determined by simple rules. Recall that a random variable X with values in I is a function $X : \Omega \rightarrow I$.

A row vector $\lambda = (\lambda_i : i \in I)$ is called a **measure** if $\lambda_i \geq 0$ for all i .

If $\sum_i \lambda_i = 1$ then it is a **distribution** (or probability measure). We start with an **initial distribution** over I , specified by $\{\lambda_i : i \in I\}$ such that $0 \leq \lambda_i \leq 1$ for all i and $\sum_{i \in I} \lambda_i = 1$.

The special case that with probability 1 we start in state i is denoted $\lambda = \delta_i = (0, \dots, 1, \dots, 0)$.

We also have a **transition matrix** $P = (p_{ij} : i, j \in I)$ with $p_{ij} \geq 0$ for all i, j .

It is a **stochastic matrix**, meaning that $p_{ij} \geq 0$ for all $i, j \in I$ and $\sum_{j \in I} p_{ij} = 1$ (i.e. each row of P is a distribution over I).

Definition 1.2. We say that $(X_n)_{n \geq 0}$ is a **Markov chain** with initial distribution λ and transition matrix P if for all $n \geq 0$ and $i_0, \dots, i_{n+1} \in I$,

$$(i) \quad P(X_0 = i_0) = \lambda_{i_0};$$

$$(ii) \quad P(X_{n+1} = i_{n+1} \mid X_0 = i_0, \dots, X_n = i_n) = P(X_{n+1} = i_{n+1} \mid X_n = i_n) = p_{i_n i_{n+1}}.$$

Markov Condition.
Is a definition. We don't have to prove it.

The future evolution of a process depends only on its current state

For short, we say $(X_n)_{n \geq 0}$ is **Markov** (λ, P) . Checking conditions (i) and (ii) is usually the most helpful way to determine whether or not a given random process $(X_n)_{n \geq 0}$ is a Markov chain. However, it can also be helpful to have the alternative description which is provided by the following theorem.

Theorem 1.3. $(X_n)_{n \geq 0}$ is Markov (λ, P) if and only if for all $n \geq 0$ and $i_0, \dots, i_n \in I$,

$$P(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}. \quad (1.1)$$

Proof. Suppose $(X_n)_{n \geq 0}$ is Markov (λ, P) . Then

$$P(X_0 = i_0, \dots, X_n = i_n)$$

$$\begin{aligned} &= P(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}) \\ &= P(X_0 = i_0) P(X_1 = i_1 \mid X_0 = i_0) \cdots P(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) \\ &= \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}. \end{aligned}$$

On the other hand if (1.1) holds we can sum it over all $i_1, \dots, i_n$, to give $P(X_0 = i_0) = \lambda_{i_0}$, i.e. (i). Then summing (1.1) on i_n we get $P(X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-2} i_{n-1}}$. Hence

$$P(X_n = i_n \mid X_0 = i_0, \dots, X_{n-1} = i_{n-1}) = \frac{P(X_0 = i_0, \dots, X_n = i_n)}{P(X_0 = i_0, \dots, X_{n-1} = i_{n-1})} = p_{i_{n-1} i_n}.$$

It's a joint probability. Any joint probability can be decomposed into a product of conditional probabilities. A joint probability is the probability that all those events happen together.

$$P(A_1, A_2, A_3) = P(A_1) P(A_2 \mid A_1) P(A_3 \mid A_2, A_1)$$

In General

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = P(X_0 = i_0) \prod_{k=1}^n p_{i_{k-1} i_k} \dots$$

$$P(X_k = i_k | X_0 = i_0, \dots, X_n = i_n)$$

which establishes (ii). $\square$

1.3 Where do Markov chains come from?

At each time we apply some new ‘randomness’ to determine the next step, in a way that is a function only of the current state. We might take $U_1, U_2, \dots$ as some i.i.d. random variables taking values in a set E and a function $F : I \times E \rightarrow I$.

Take $i \in I$. Set $X_0 = i$ and define recursively $X_{n+1} = F(X_n, U_{n+1})$, $n \geq 0$. Then $(X_n)_{n \geq 0}$ is *Markov* (δ_i, P) where $p_{ij} = P(F(i, U) = j)$.

1.4 How can we simulate them?

Use a computer to simulate $U_1, U_2, \dots$ as i.i.d. $U[0, 1]$. Define $F(U, i) = J$ by the rules

$$\begin{aligned} U \in [0, p_{i1}) &\implies J = 1 \\ U \in \left[\sum_{k=1}^{j-1} p_{ik}, \sum_{k=1}^j p_{ik} \right), \quad j \geq 2 &\implies J = j. \end{aligned}$$

1.5 The n -step transition matrix

Let A be an event. A convenient notation is $P_i(A) = P(A | X_0 = i)$. For example

$$P_i(X_1 = j) = p_{ij}.$$

Given the initial distribution λ , let us treat it as a row vector. Then

$$P(X_1 = j) = \sum_{i \in I} \lambda_i P_i(X_1 = j) = \sum_{i \in I} \lambda_i p_{ij}.$$

Similarly,

$$\begin{aligned} P_i(X_2 = j) &= \sum_k P_i(X_1 = k, X_2 = j) = \sum_k p_{ik} p_{kj} = (P^2)_{ij} \\ P(X_2 = j) &= \sum_{i,k} \lambda_i P_i(X_1 = k, X_2 = j) = \sum_{i,k} \lambda_i p_{ik} p_{kj} = (\lambda P^2)_j. \end{aligned}$$

Continuing in this way,

$$\begin{aligned} P_i(X_n = j) &= (\delta_i P^n)_j = (P^n)_{ij} = p_{ij}^{(n)} \\ P(X_n = j) &= \sum_{i_0, \dots, i_{n-1}} \lambda_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} j} = (\lambda P^n)_j. \end{aligned}$$

Thus $P^{(n)} = (p_{ij}^{(n)})$, the n -step transition matrix, is simply P^n (P raised to power n).

Also, for all i, j and $n, m \geq 0$, the (obvious) **Chapman-Kolmogorov equations** hold:

$$p_{ij}^{(n+m)} = \sum_{k \in I} p_{ik}^{(n)} p_{kj}^{(m)}$$

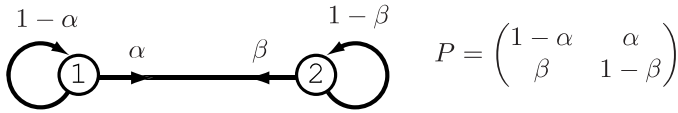
named for their independent formulation by Chapman (a Trinity College graduate, 1880-1970), and Kolmogorov (1903-1987).

In Example 1.1

$$P^{(n)} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}^n \rightarrow \begin{pmatrix} \frac{1}{5} & \frac{2}{5} & \frac{2}{5} & 0 & 0 & 0 & 0 \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} & 0 & 0 & 0 & 0 \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} & 0 & 0 & 0 & 0 \\ \frac{2}{15} & \frac{4}{15} & \frac{4}{15} & 0 & 0 & 0 & \frac{1}{3} \\ \frac{1}{15} & \frac{2}{15} & \frac{2}{15} & 0 & 0 & 0 & \frac{2}{3} \\ \frac{2}{15} & \frac{4}{15} & \frac{4}{15} & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

1.6 $P^{(n)}$ for a two-state Markov chain

Example 1.4 (A two-state Markov chain).



The eigenvalues are 1 and $1 - \alpha - \beta$. So we can write

$$P = U \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha - \beta \end{pmatrix} U^{-1} \implies P^n = U \begin{pmatrix} 1 & 0 \\ 0 & (1 - \alpha - \beta)^n \end{pmatrix} U^{-1}$$

So $p_{11}^{(n)} = A + B(1 - \alpha - \beta)^n$ for some A and B .

But $p_{11}^{(0)} = 1 = A + B$ and $p_{11}^{(1)} = 1 - \alpha = A + B(1 - \alpha - \beta)$. So $(A, B) = (\beta, \alpha)/(\alpha + \beta)$, i.e.

$$p_{11}^{(n)} = \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta}(1 - \alpha - \beta)^n.$$

Note that this tends exponentially fast to a limit of $\beta/(\alpha + \beta)$.

Other components of P^n can be computed similarly, and we have

$$P^n = \begin{pmatrix} \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta}(1 - \alpha - \beta)^n & \frac{\alpha}{\alpha + \beta} - \frac{\alpha}{\alpha + \beta}(1 - \alpha - \beta)^n \\ \frac{\beta}{\alpha + \beta} - \frac{\beta}{\alpha + \beta}(1 - \alpha - \beta)^n & \frac{\alpha}{\alpha + \beta} + \frac{\beta}{\alpha + \beta}(1 - \alpha - \beta)^n \end{pmatrix}$$

Note. We might have reached the same answer by arguing:

$$p_{11}^{(n)} = p_{11}^{(n-1)}p_{11} + p_{12}^{(n-1)}p_{21} = p_{11}^{(n-1)}(1 - \alpha) + (1 - p_{11}^{(n-1)})\beta.$$

This gives the linear recurrence relation

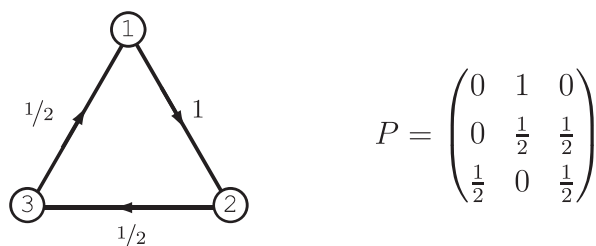
$$p_{11}^{(n)} = \beta + (1 - \alpha - \beta)p_{11}^{(n-1)}$$

to which the solution is of the form $p_{11}^{(n)} = A + B(1 - \alpha - \beta)^n$.

2 Calculation of n -step transition probabilities, class structure, absorption, and irreducibility

2.1 Example: a three-state Markov chain

Example 2.1.



Now $0 = \det(xI - P) = x(x - 1/2)^2 - 1/4 = (1/4)(x - 1)(4x^2 + 1)$.
Eigenvalues are $1, \pm i/2$. This means that

$$p_{11}^{(n)} = A + B(i/2)^n + C(-i/2)^n.$$

We make the substitution:

$$(\pm i/2)^n = (1/2)^n e^{\pm i n \pi / 2} = (1/2)^n \left(\cos(n\pi/2) \pm i \sin(n\pi/2) \right).$$

Thus for some B' and C' ,

$$p_{11}^{(n)} = A + (1/2)^n \left(B' \cos(n\pi/2) + C' \sin(n\pi/2) \right).$$

We then use facts that $p_{11}^{(0)} = 1$, $p_{11}^{(1)} = 0$, $p_{11}^{(2)} = 0$ to fix A , B' and C' and get

$$p_{11}^{(n)} = \frac{1}{5} + \left(\frac{1}{2}\right)^n \left[\frac{4}{5} \cos \frac{n\pi}{2} - \frac{2}{5} \sin \frac{n\pi}{2} \right].$$

Note that the second term is exponentially decaying.

We have $p_{11}^{(5)} = \frac{3}{16}$, $p_{11}^{(10)} = \frac{51}{256}$, $|p_{11}^{(n)} - 1/5| < 2^{-n}$.

More generally, for chain with m states, and states i and j

- (i) Compute the eigenvalues $\mu_1, \dots, \mu_m$ of the $m \times m$ matrix P .
- (ii) If the eigenvalues are distinct then $p_{ij}^{(n)}$ has the form (remembering that $\mu_1 = 1$),

$$p_{ij}^{(n)} = a_1 + a_2 \mu_2^n + \dots + a_m \mu_m^n$$

for some constants $a_1, \dots, a_m$. If an eigenvalue μ is repeated k times then there will be a term of $(b_0 + b_1 n + \dots + b_{k-1} n^{k-1}) \mu^n$.

- (iii) Complex eigenvalues come in conjugate pairs and these can be written using sines and cosines, as in the example.

However, sometimes there is a more clever way.

2.2 Example: use of symmetry

Example 2.2 (Random walk on vertices of a complete graph). Consider a random walk on the complete graph K_4 (vertices of a tetrahedron), with

$$P = \frac{1}{3} \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}.$$

Eigenvalues are $\{1, -1/3, -1/3, -1/3\}$ so general solution is $p_{11}^{(n)} = A + (-1/3)^n(a + bn + cn^2)$. However, we may use symmetry in a helpful way. Notice that for $i \neq j$, $p_{ij}^{(n)} = (1/3)(1 - p_{ii}^{(n)})$. So

$$p_{11}^{(n)} = \sum_{j \neq 1} p_{1j} p_{j1}^{(n-1)} = (1/3) \left(1 - p_{11}^{(n-1)}\right).$$

Thus we have just a first order recurrence equation, to which the general solution is of the form $p_{11}^{(n)} = A + B(-1/3)^n$. Since $A + B = 1$ and $A + (-1/3)B = 0$ we have $p_{11}^{(n)} = 1/4 + (3/4)(-1/3)^n$. Obviously, $p_{11}^{(n)} \rightarrow 1/4$ as $n \rightarrow \infty$.

Do we expect the same sort of thing for random walk on corners of a cube? (No, $p_{11}^{(n)}$ does not tend to a limit since $p_{11}^{(n)} = 0$ if n is odd.)

2.3 Markov property

Theorem 2.3 (Markov property). *Let $(X_n)_{n \geq 0}$ be Markov(λ, P). Then conditional on $X_m = i$, $(X_{m+n})_{n \geq 0}$ is Markov(δ_i, P) and is independent of the random variables $X_0, X_1, \dots, X_m$.*

Proof (non-examinable). We must show that for any event A determined by $X_0, \dots, X_m$ we have

$$\begin{aligned} P(\{X_m = i_m, \dots, X_{m+n} = i_{m+n}\} \cap A \mid X_m = i) \\ = \delta_{ii_m} p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}} P(A \mid X_m = i) \end{aligned} \quad (2.1)$$

then the result follows from Theorem 1.3. First consider the case of elementary events like $A = \{X_0 = i_0, \dots, X_m = i_m\}$. In that case we have to show that

$$\begin{aligned} P(X_0 = i_0, \dots, X_{m+n} = i_{m+n} \text{ and } i = i_m) / P(X_m = i) \\ = \delta_{ii_m} p_{i_m i_{m+1}} \cdots p_{i_{m+n-1} i_{m+n}} P(X_0 = i_0, \dots, X_m = i_m \text{ and } i = i_m) / P(X_m = i) \end{aligned}$$

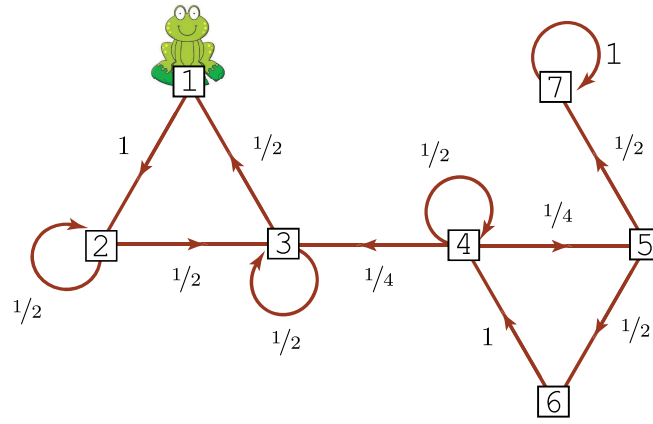
which is true by Theorem 1.3. In general, any event A determined by $X_0, \dots, X_m$ may be written as the union of a countable number of disjoint elementary events

$$A = \bigcup_{k=1}^{\infty} A_k.$$

The desired identity (2.1) follows by summing the corresponding identities for the A_k . $\square$

2.4 Class structure

It is sometimes possible to break a Markov chain into smaller pieces, each of which is relatively easy to understand, and which together give an understanding of the whole. This is done by identifying communicating classes of the chain. Recall the initial example.



We say that i **leads to** j and write $i \rightarrow j$ if

$$P_i(X_n = j \text{ for some } n \geq 0) > 0.$$

We say i **communicates with** j and write $i \leftrightarrow j$ if both $i \rightarrow j$ and $j \rightarrow i$.

Theorem 2.4. For distinct states i and j the following are equivalent.

- (i) $i \rightarrow j$;
- (ii) $p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} > 0$ for some states $i_0, i_1, \dots, i_n$, where $n \geq 1$, $i_0 = i$ and $i_n = j$;
- (iii) $p_{ij}^{(n)} > 0$ for some $n \geq 1$.

Proof. Observe that

$$p_{ij}^{(n)} \leq P_i(X_n = j \text{ for some } n \geq 0) \leq \sum_{n=0}^{\infty} p_{ij}^{(n)}$$

which proves the equivalence of (i) and (iii). Also,

$$p_{ij}^{(n)} = \sum_{i_1, \dots, i_{n-1}} p_{i_0 i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n}$$

so that (ii) and (iii) are equivalent. $\square$

2.5 Closed classes

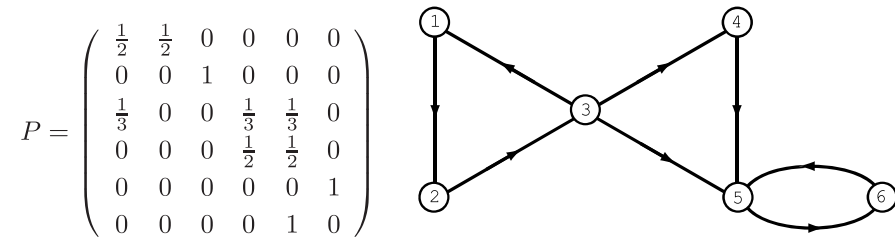
It is clear from (ii) that $i \leftrightarrow j$ and $j \leftrightarrow k$ imply $i \leftrightarrow k$, and also that $i \leftrightarrow i$. So $\leftrightarrow$ satisfies the conditions for an equivalence relation on I and so partitions I into **communicating classes**. We say a C is a **closed class** if

$$i \in C, i \rightarrow j \implies j \in C.$$

A closed class is one from which there is no escape. A state i is **absorbing** if $\{i\}$ is a closed class.

If C is not closed then it is **open** and there exist $i \in C$ and $j \notin C$ with $i \rightarrow j$ (you can escape).

Example 2.5. Find the classes in P and say whether they are open or closed.



The solution is obvious from the diagram. The classes are $\{1, 2, 3\}$, $\{4\}$ and $\{5, 6\}$, with only $\{5, 6\}$ being closed.

2.6 Irreducibility

A chain or transition matrix P in which I is a single class is called **irreducible**. It is easy to detect. We just have to check that $i \rightarrow j$ for every i, j .

3 Recurrent and transient states

Definition 3.1 A state i is called **recurrent** if

$$\mathbb{P} \left(\bigcup_{n=1}^{\infty} \{X_n = i\} \mid \{X_0 = i\} \right) = 1,$$

otherwise it is called **transient**.

According to the above definition i is recurrent if, starting from the state i at time 0 the Markov chain returns to the state i almost surely in a finite time. In particular if $p_{ii} = 1$, then i is called **absorbing** or **trap**.

Definition 3.2 Let j be a state. We call **first entrance time in j** (or **first visit time in j**) the random variable with values $\mathbb{N} \cup \{+\infty\}$

$$T_j(\omega) = \begin{cases} \min\{n \geq 1 \mid X_n(\omega) = j\} & \text{se } \{n \geq 1 \mid X_n(\omega) = j\} \neq \emptyset, \\ +\infty & \text{altrimenti.} \end{cases}$$

Note that, for all state j and all $n > 1$, we have

$$\begin{aligned} \{T_j < +\infty\} &= \bigcup_{1 \leq n < +\infty} \{X_n = j\}, \\ \{T_j = 1\} &= \{X_1 = j\}, \\ \{T_j = n\} &= \{X_n = j\} \cap \bigcap_{1 \leq m \leq n-1} \{X_m \neq j\}. \end{aligned}$$

Notation. For simplicity we denote by $\mathbb{P}_i$ the conditional probability

$$\mathbb{P}_i(\cdot) = \mathbb{P}(\cdot \mid \{X_0 = i\}).$$

For all $n > 0$ and all pair $i, j \in I$ let

$$f_{ij}^{(n)} = \mathbb{P}_i\{T_j = n\}, \quad f_{ij}^{(0)} = 0.$$

and also

$$f_{ij}^* = \sum_{1 \leq n < +\infty} f_{ij}^{(n)} = \mathbb{P}_i\{T_j < +\infty\}.$$

The Markov $(X_n)_{n \geq 0}$ is time-homogeneous therefore

$$f_{ij}^{(n)} = \mathbb{P}(\{X_{m+n} = j\} \cap \bigcap_{1 \leq \nu \leq n-1} \{X_{m+\nu} \neq j\} \mid \{X_m = i\}).$$

Note that a state i is recurrent if $f_{ii}^* = 1$ and transient if $f_{ii}^* < 1$.

Theorem 3.3 (Renewal equation) *Let $i, j \in I$. For all $n \geq 1$ we have*

$$p_{ij}^{(n)} = \sum_{\nu=1}^n f_{ij}^{(\nu)} p_{jj}^{(n-\nu)}. \quad (6)$$

PROOF. Recall that $\mathbb{P}_i := \mathbb{P}\{\cdot \mid \{X_0 = i\}\}$ by definition. Since

$$\{X_n = j\} = \bigcup_{\nu=1}^n (\{X_n = j\} \cap \{T_j = \nu\})$$

by the Markov property

$$\begin{aligned} p_{ij}^{(n)} &= \mathbb{P}\{X_n = j\} \\ &= \sum_{1 \leq \nu \leq n} \mathbb{P}\{X_n = j, T_j = \nu\} \\ &= \mathbb{P}\{X_n = j, X_{n-1} \neq j, \dots, X_2 \neq j, X_1 \neq j\} \\ &\quad + \sum_{1 \leq \nu \leq n-1} \mathbb{P}(\{X_n = j, X_\nu = j, X_{\nu-1} \neq j, \dots, X_1 \neq j\}) \\ &= f_{ij}^{(n)} + \sum_{1 \leq \nu \leq n-1} \mathbb{P}\{X_n = j \mid X_\nu = j, X_{\nu-1} \neq j, \dots, X_1 \neq j\} \\ &\quad \cdot \mathbb{P}\{X_\nu = j, X_{\nu-1} \neq j, \dots, X_1 \neq j\} \\ &= f_{ij}^{(n)} + \sum_{1 \leq \nu \leq n-1} \mathbb{P}\{X_n = j \mid X_\nu = j\} \mathbb{P}\{T_j = \nu\} \\ &= f_{ij}^{(n)} + \sum_{1 \leq \nu \leq n-1} f_{ij}^{(\nu)} p_{jj}^{(n-\nu)}. \end{aligned}$$

This proves (6). ■

In the proof of the following Theorem 3.5 we need the following Lemma (also known as Abel lemma) that could be proved directly. However, it is an immediate consequence of the monotone convergence theorem, therefore we skip the proof.

Lemma 3.4 *Let $(a_n)_{n \geq 0}$ be a sequence of non-negative real numbers such that the power series*

$$A(s) = \sum_{n=0}^{\infty} a_n s^n$$

has convergence radius $r \geq 1$. Then

$$\lim_{s \rightarrow 1^-} A(s) = \sum_{n=0}^{\infty} a_n.$$

Theorem 3.5 *For all state i the following are equivalent:*

- a) *The state i is recurrent,*
- b) *The series $\sum_{n=0}^{\infty} p_{ii}^{(n)}$ is divergent.*

If i is transient then

$$\sum_{n=0}^{\infty} p_{ii}^{(n)} = \frac{1}{1 - f_{ii}^*} = \frac{1}{\mathbb{P}_i\{T_j = \infty\}}$$

PROOF. Consider generating functions of sequences $(p_{ii}^{(n)})_{n \geq 0}$, $(f_{ii}^{(n)})_{n \geq 0}$

$$P_{ii}(s) = \sum_{n=0}^{\infty} p_{ii}^{(n)} s^n, \quad F_{ii}(s) = \sum_{n=0}^{\infty} f_{ii}^{(n)} s^n.$$

Note that in both cases the convergence radius is $r \geq 1$ because sequences $(p_{ii}^{(n)})_{n \geq 0}$ and $(f_{ii}^{(n)})_{n \geq 0}$ are bounded. By Theorem 3.3 (renewal equation), for all $n \geq 1$ we have

$$p_{ii}^{(n)} s^n = s^n \sum_{\nu=1}^n f_{ii}^{(\nu)} p_{ii}^{(n-\nu)} = \sum_{\nu=1}^n f_{ii}^{(\nu)} s^{\nu} p_{ii}^{(n-\nu)} s^{n-\nu}$$

and so

$$\begin{aligned} P_{ii}(s) - 1 &= \sum_{n=1}^{\infty} p_{ii}^{(n)} s^n \\ &= \sum_{n=1}^{\infty} \sum_{\nu=1}^n f_{ii}^{(\nu)} s^{\nu} p_{ii}^{(n-\nu)} s^{n-\nu} \\ &= \sum_{\nu=1}^{\infty} f_{ii}^{(\nu)} s^{\nu} \sum_{n=\nu}^{\infty} p_{ii}^{(n-\nu)} s^{n-\nu} \\ &= \sum_{\nu=1}^{\infty} f_{ii}^{(\nu)} s^{\nu} \sum_{n=0}^{\infty} p_{ii}^{(n)} s^n \\ &= F_{ii}(s) P_{ii}(s). \end{aligned}$$

Therefore, for all $s \in (-1, 1)$,

$$P_{ii}(s) = \frac{1}{1 - F_{ii}(s)}.$$

The conclusion follows from an application of Abel's lemma noting that

$$\lim_{s \rightarrow 1^-} F_{ii}(s) = f_{ii}^*, \quad \lim_{s \rightarrow 1^-} P_{ii}(s) = \sum_{n=0}^{\infty} p_{ii}^{(n)}$$

■

Corollary 3.6 *Let J be a class of states and $j \in J$. If j is recurrent (respectively, transient) then all elements of J are recurrent (respectively, transient).*

PROOF. Let $k \in J - \{j\}$. Since k and j communicate, there exists $l, m \geq 1$ such that

$$p_{jk}^{(l)} > 0, \quad p_{kj}^{(m)} > 0.$$

For all integer $n \geq 1$ we have

$$\begin{aligned} p_{kk}^{(l+n+m)} &\geq p_{kj}^{(m)} p_{jj}^{(n)} p_{jk}^{(l)} \\ p_{jj}^{(l+n+m)} &\geq p_{jk}^{(l)} p_{kk}^{(n)} p_{kj}^{(m)} \end{aligned}$$

so that series

$$\sum_{n=0}^{\infty} p_{kk}^{(n)}, \quad \sum_{n=0}^{\infty} p_{jj}^{(n)}$$

are both convergent or both divergent. The conclusion follows from Theorem 3.5. $\blacksquare$

Notice that, for all state j , the sum of the series

$$\sum_{n \geq 0} p_{ij}^{(n)}$$

is the average sojourn time in j (average number of visits to j) starting from i . Indeed the random variable

$$\sum_{n \geq 0} \mathbb{1}_{\{X_n = j\}}$$

represents the random time spent in j because it counts how many times we observe the event $\{X_n = j\}$ and his expectation is

$$\mathbb{E}_i \left[\sum_{n \geq 0} \mathbb{1}_{\{X_n = j\}} \right] = \sum_{n \geq 0} \mathbb{E}_i [\mathbb{1}_{\{X_n = j\}}] = \sum_{n \geq 0} \mathbb{P}\{X_n = j\} = \sum_{n \geq 0} p_{ij}^{(n)}. \quad (7)$$

Corollary 3.7 *If j is a transient state then, for all state i the series*

$$\sum_{n=0}^{\infty} p_{ij}^{(n)} \quad (8)$$

is convergent. In particular, starting from any state i , the Markov chain visits j only a finite number of times almost surely.

PROOF. If $i = j$ convergence follows from Theorem 3.5. If $i \neq j$ then, by the renewal equation (6), we have

$$\sum_{n=1}^{\infty} p_{ij}^{(n)} = \sum_{n=1}^{\infty} \sum_{\nu=1}^n f_{ij}^{(\nu)} p_{jj}^{(n-\nu)}$$

$$\begin{aligned}
&= \sum_{\nu=1}^{\infty} f_{ij}^{(\nu)} \sum_{n=\nu}^{\infty} p_{jj}^{(n-\nu)} \\
&= f_{ij}^* \sum_{n=0}^{\infty} p_{jj}^{(n)} < \infty.
\end{aligned}$$

In particular, by (7), the random variable

$$\sum_{n \geq 0} \mathbb{1}_{\{X_n = j\}}$$

has finite expectation. It follows that it is almost surely finite and so only a finite number of events $\{X_n = j\}$ occurs. ■

In the case where the number of states is finite the Markov chain cannot visit only transient states (each one a finite number of times). As a consequence, we have the following

Corollary 3.8 *A Markov chain with a finite set of states I has at least a recurrent state.*

PROOF. Suppose by contradiction that all states are transient. For all i, j , the series (8) is convergent, in particular

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = 0.$$

The matrix $p^{(n)}$ is stochastic and so, for all $n \geq 1$, we have also

$$\sum_{j \in I} p_{ij}^{(n)} = 1.$$

Taking the limit as n goes to infinity and exchanging limit and sum on j we find the contradiction $0 = 1$. ■

4 Random Walks on $\mathbb{Z}^d$

We apply Theorem 3.5 to the study of random walks on $\mathbb{Z}^d$.

4.1 Random Walk on $\mathbb{Z}$

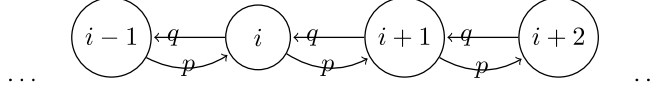
Let $(X_n)_{n \geq 0}$ be a Markov chain with set of states $\mathbb{Z}$ and transition matrix

$$P = \begin{pmatrix} \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & q & 0 & p & 0 & 0 & \dots \\ \dots & 0 & q & 0 & p & 0 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$

where $p, q \in \mathbb{R}$ and

$$p, q \in]0, 1[, \quad p + q = 1.$$

Clearly $(X_n)_{n \geq 0}$ is irreducible and all states have period 2.



We want to establish if states are recurrent or transient. For all $n \geq 0$, compute

$$P_{00}^{(n)} = \binom{2n}{n} (pq)^n.$$

By Stirling's formula

$$\lim_{n \rightarrow \infty} \frac{n^n \exp(-n) \sqrt{2\pi n}}{n!} = 1$$

we find the asymptotic behaviour

$$P_{00}^{(n)} \approx \frac{(4pq)^n}{\sqrt{\pi n}}.$$

Since $0 \leq 4pq = 4p(1-p) \leq 1$ and $4pq = 1$ if and only if $p = q = 1/2$ and $4pq < 1$ otherwise, by Theorem 3.5 we deduce that

Proposition 4.1 *The following hold:*

- *Symmetric random walk ($p = q = 1/2$): all states are recurrent.*
- *Asymmetric random walk ($p \neq q$): all states are transient.*

In the case where $p = q = 1/2$ first return times T_i to a state i are almost surely finite. However, by means of generating functions F_{00} e P_{00} introduced in the proof of Theorem 3.5 we can show that their expectation $\mathbb{E}_i[T_i]$ is infinite. In other words, return to i is almost sure but it will occur (on average) after a very long time.

Let us begin by recalling the Taylor series expansion,

$$(1 - 4x)^{-\frac{1}{2}} = \sum_{n \geq 0} \binom{2n}{n} x^n, \quad |x| < \frac{1}{4},$$

that gives an explicit formula for the generating function P_{00}

$$P_{00}(s) = \frac{1}{\sqrt{1 - 4pqs^2}}, \quad |s| < 1.$$

The generating function F_{00} is

$$F_{00}(s) = 1 - (P_{00}(s))^{-1} = 1 - \sqrt{1 - 4pqs^2}, \quad |s| < 1.$$

By Abel's Lemma, since $4pq = 1$, we finally find

$$\begin{aligned} \mathbb{E}_0[T_0] &= \sum_{n \geq 1} n f_{00}^{(n)} = \lim_{s \rightarrow 1^-} \sum_{n \geq 1} n f_{00}^{(n)} s^{n-1} \\ &= \lim_{s \rightarrow 1^-} \frac{dF_{00}(s)}{ds} = \lim_{s \rightarrow 1^-} \frac{4pqs}{\sqrt{1 - 4pqs^2}} = +\infty. \end{aligned}$$

4.2 Random walk on $\mathbb{Z}^2$

Consider a Markov chain $(X_n)_{n \geq 0}$ with set of states $\mathbb{Z}^2$ and transition probabilities

$$\mathbb{P}\{X_{n+1} = (i, j) \mid X_n = (h, k)\} = \begin{cases} 1/4 & \text{se } |i - h| + |j - k| = 1, \\ 0 & \text{altrimenti.} \end{cases}$$

$(X_n)_{n \geq 0}$ is irreducible and has period 2. Let P be the transition matrix

$$P_{00}^{(2n)} = \sum_{h+k=n} \frac{(2n)!}{(h!)^2(k!)^2} \left(\frac{1}{4}\right)^{2n}$$

(where 0 denotes $(0, 0) \in \mathbb{Z}^2$). Simple computations yield

$$\begin{aligned} P_{00}^{(2n)} &= \sum_{k=0}^n \frac{(2n)!}{(k!)^2((n-k)!)^2} \left(\frac{1}{4}\right)^{2n} \\ &= \binom{2n}{n} \left(\frac{1}{4}\right)^{2n} \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} \end{aligned}$$

Comparing terms $a^n b^n$ in the Newton binomial of $(a+b)^{2n}$ and the square of the Newton binomial of $(a+b)^n$ we find the remarkable formula

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k}.$$

Summing up

$$P_{00}^{2n} = \left(\binom{2n}{n} \left(\frac{1}{4}\right)^n \right)^2,$$

and, by Stirling's formula,

$$P_{00}^{2n} \approx \frac{1}{\pi n}$$

allowing us to prove that, by Theorem 3.5 and Corollary 3.6 all states are recurrent.

4.3 Random walk on $\mathbb{Z}^d$ for $d \geq 3$

Transition probabilities of the Markov chain $(X_n)_{n \geq 0}$ are now

$$\mathbb{P}\{X_{n+1} = (j_1, \dots, j_d) \mid X_n = (i_1, \dots, i_d)\} = \begin{cases} (2d)^{-1} & \text{if } \sum_{k=1}^d |i_k - j_k| = 1, \\ 0 & \text{otherwise.} \end{cases}$$

In analogy with random walks on $\mathbb{Z}$ and $\mathbb{Z}^2$ we have

$$\begin{aligned} P_{00}^{(2n)} &= \sum_{k_1 + \dots + k_d = n} \frac{(2n)!}{(k_1!)^2 \dots (k_d!)^2} \left(\frac{1}{2d}\right)^{2n} \\ &= \binom{2n}{n} \left(\frac{1}{2d}\right)^{2n} \sum_{k_1 + \dots + k_d = n} \frac{(n!)^2}{(k_1!)^2 \dots (k_d!)^2}. \end{aligned}$$

The above sum is dominated by

$$\max_{k_1+\dots+k_d=n} \left\{ \frac{n!}{k_1! \dots k_d!} \right\} \sum_{k_1+\dots+k_d=n} \frac{n!}{k_1! \dots k_d!} = d^n \max_{k_1+\dots+k_d=n} \left\{ \frac{n!}{k_1! \dots k_d!} \right\}$$

where we used a simple extension of Newton's binomial formula

$$\sum_{k_1+\dots+k_d=n} \frac{n!}{k_1! \dots k_d!} x_1^{k_1} \dots x_d^{k_d} = (x_1 + \dots + x_d)^n.$$

Elementary computations show that the maximum value is attained for

$$k_1 = k_2 = \dots = k_d = \frac{n}{d}.$$

By Stirling's formula we find that $(P_{00}^{(2n)})_{n \geq 0}$ is asymptotically smaller than a constant times

$$\binom{2n}{n} \left(\frac{1}{2d} \right)^{2n} \frac{n! d^n}{((n/d)!)^d} \approx \frac{d^{d/2}}{\pi^{d/2} 2^{(d-1)/2}} \frac{1}{n^{d/2}}.$$

Since $d \geq 3$, states are transient because $\sum_{n \geq 0} P_{00}^{(2n)} < +\infty$.

5 Invariant densities and limit theorems

Let $(X_n)_{n \geq 0}$ be a Markov chain on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with set of states I and transition matrix $P = (p_{ij})_{i,j \in I}$.

Definition 5.1 *A probability measure μ on I is called invariant for the Markov chain $(X_n)_{n \geq 0}$ if the random variable X_n has distribution μ for all $n \geq 0$.*

We identify a probability measure on I with its density (with respect to the counting measure on the σ -algebra of all subsets of I) namely the collection of numbers $(\mu_i)_{i \in I}$ in the interval $[0, 1]$ such that $\sum_{i \in I} \mu_i = 1$.

Proposition 5.2 *A probability measure μ on I is invariant for the Markov chain $(X_n)_{n \geq 0}$ if and only if, for all state i , we have*

$$\mu_i = \sum_{j \in I} \mu_j p_{ji} \quad (9)$$

(i.e. its density is a left eigenvalue for the transition probability matrix with eigenvalue 1).

PROOF. Let μ be the probability distribution of X_0 , the probability distribution ν of X_1 has density

$$\nu_i = \mathbb{P}\{X_1 = i\} = \sum_{j \in I} \mathbb{P}\{X_0 = j\} p_{ji} = \sum_{j \in I} \mu_j p_{ji},$$

Therefore, if μ is an invariant probability measure for the Markov chain $(X_n)_{n \geq 0}$, then (9) holds. Conversely, if μ is a probability measure on I satisfying (9), one can easily show that X_n has probability distribution μ for all n . ■

Summing up a probability measure μ on I , identifying its density with the vector $(\mu_i)_{i \in I}$, is invariant if and only if

$$\mu = \mu P$$

where P is the transition matrix.

Theorem 5.3 *If the set of states I is finite, then there exists at least an invariant probability measure.*

PROOF. The set of probability densities on I

$$\mathcal{K} = \left\{ (\mu_i)_{i \in I} \mid 0 \leq \mu_i \leq 1, \text{ per ogni } i, \sum_{i \in I} \mu_i = 1 \right\}$$

is compact as a closed bounded subset of $\mathbb{R}^{\text{card}(I)}$. Let $\mu^{(0)} \in \mathcal{K}$. It is easy to show that, for all $n \geq 1$, the family of real numbers $(\mu_j^{(n)})_{j \in I}$ defined by

$$\mu_j^{(n)} = \frac{1}{n} \sum_{k=1}^n \left(\sum_{i \in I} \mu_i^{(0)} p_{ij}^{(k)} \right)$$

is a probability density on I . Since the set $\mathcal{K}$ is compact, by considering a subsequence, if necessary, we can assume that the sequence $(\mu_j^{(n)})_{n \geq 0}$ is convergent for all $j \in I$. Defining

$$\pi_j = \lim_{n \rightarrow \infty} \mu_j^{(n)}$$

the collection of real numbers $(\pi_j)_{j \in I}$ is a probability density on I and, moreover,

$$\begin{aligned} \sum_{i \in I} \pi_i p_{ij} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \left(\sum_{i \in I} \pi_i p_{ij}^{(k+1)} \right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=1}^{n+1} \left(\sum_{i \in I} \pi_i p_{ij}^{(k)} \right) - \lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{i \in I} \pi_i p_{ij} \\ &= \pi_j \end{aligned}$$

for all $j \in I$. This proves that the probability measure with density $(\pi_i)_{i \in I}$ is invariant for the Markov chain. $\blacksquare$

It is worth noticing that, in the above proof, the invariant probability density has been obtained as a limit of subsequences of $(\mu^{(n)})_{n \geq 1}$ where $\mu^{(0)}$ is a density on I and

$$\mu_j^{(n)} = \frac{1}{n} \sum_{k=1}^n (\mu^{(0)} P^k)_j$$

for all $n \geq 1$ and all $j \in I$ (here $(\mu^{(0)} P^k)_j = \sum_{i \in I} \mu_i^{(0)} p_{ij}^{(k)}$). In particular, if $\mu^{(0)} = \delta_i$ (Dirac density in i), we have

$$\mu_j^{(n)} = \frac{1}{n} \sum_{k=1}^n p_{ij}^{(k)}.$$

The analysis of the above formula allows us to interpret the meaning of an invariant distribution. Indeed, recalling that the random variable

$$\sum_{k=1}^n \mathbb{1}_{\{X_k=j\}}$$

represents the number of visits in j up to time n , it is clear that the random variable

$$\frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k=j\}}$$

represents the random *frequency* of visits in the state j at time n . Thus its expectation, with respect to the probability $\mathbb{P}_i$,

$$\mathbb{E}_i \left[\frac{1}{n} \sum_{k=1}^n \mathbb{1}_{\{X_k=j\}} \right] = \frac{1}{n} \sum_{k=1}^n \mathbb{P}_i \{X_k = j\} = \frac{1}{n} \sum_{k=1}^n p_{ij}^{(k)}$$

represents *average frequency* of visits in the state j at time n . Therefore we can interpret values π_j as *asymptotic average frequencies* of visits in the state j .

A Markov chain (also with I finite) may have more than an invariant density as shows the example of *Gambler's ruin* Markov chain with states $\{0, 1, \dots, N\}$. In this case the densities

$$(1, 0, \dots, 0), \quad (0, \dots, 0, 1)$$

and all their convex combinations

$$\lambda(1, 0, \dots, 0) + (1 - \lambda)(0, \dots, 0, 1) \quad \lambda \in [0, 1]$$

are invariant.

In the case where the set I is infinite, however, a Markov chain may not admit any invariant density as shows the following

Example 5.1 The *Random walk on $\mathbb{Z}$* has no invariant density. Indeed, such an invariant density $(\pi_i)_{i \in \mathbb{Z}}$ would satisfy

$$\pi_i = q\pi_{i+1} + p\pi_{i-1}$$

for all $i \in \mathbb{Z}$, namely

$$q(\pi_{i+1} - \pi_i) = p(\pi_i - \pi_{i-1}). \quad (10)$$

We begin by observing that $\pi_{i+1} \neq \pi_i$ for all $i \in \mathbb{Z}$ because, otherwise, the above identity implies $\pi_{i+1} = \pi_i$ for all i and $(\pi_i)_{i \in \mathbb{Z}}$ cannot be a probability density.

We distinguish three cases $p > q$, $p = q = 1/2$ e $p < q$. In the first case from (10) we easily find

$$\pi_{i+1} - \pi_i = (p/q)^i (\pi_1 - \pi_0)$$

from which

$$\lim_{i \rightarrow +\infty} |\pi_{i+1} - \pi_i| = +\infty$$

contradicting boundedness of $(\pi_i)_{i \in \mathbb{Z}}$. In the second case from (10) we get the formula

$$\pi_{i+1} - \pi_i = \pi_1 - \pi_0$$

from which

$$\pi_i = i(\pi_1 - \pi_0) + c$$

(with c constant) that again contradicts boundedness of $(\pi_i)_{i \in \mathbb{Z}}$. The third case is similar to the first one.

Summing up, the random walk on $\mathbb{Z}$ has no invariant density.

The explicit computation or just the proof of the existence of an invariant density might not be simple.

The following Markov chain, that arises in several models (Gambler's ruin problem when the gambler's opponent has an infinite fortune, e.g. a bank, a random walk on $\mathbb{N}$, the number of customers in a queue at a counter, ...) passeggiata casuale su $\mathbb{N}$, ...) is a remarkable case in which one can compute explicitly the unique invariant density.

Example 5.2 Consider the Markov chain with set of states $\mathbb{N}$ and transition matrix

$$\begin{pmatrix} r_0 & p_0 & 0 & 0 & 0 & \dots \\ q_1 & r_1 & p_1 & 0 & 0 & \dots \\ 0 & q_2 & r_2 & p_2 & 0 & \dots \\ 0 & 0 & q_3 & r_3 & p_3 & \dots \\ 0 & 0 & 0 & q_4 & r_4 & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$

where $(p_m)_{m \geq 0}$, $(r_m)_{m \geq 0}$, $(q_m)_{m > 0}$ are three sequences of random numbers with $p_m, q_m \in]0, 1[$ and $r_m \in [0, 1]$ such that

$$r_0 + p_0 = 1, \quad q_m + r_m + p_m = 1$$

for all $m \geq 1$. Every invariant density $(\pi_n)_{n \geq 0}$ satisfies

$$\begin{aligned} \pi_0 &= r_0 \pi_0 + q_1 \pi_1 \\ \pi_n &= p_{n-1} \pi_{n-1} + r_n \pi_n + q_{n+1} \pi_{n+1} \end{aligned}$$

for all $n \geq 1$ besides positivity $\pi_n \geq 0$ and normalization

$$\sum_{n \geq 0} \pi_n = 1.$$

From the first equation we find

$$\pi_1 = \frac{1 - r_0}{q_1} \pi_0 = \frac{p_0}{q_1} \pi_0$$

and, inductively,

$$\pi_n = \frac{p_0 \cdots p_{n-1}}{q_1 \cdots q_n} \pi_0$$

for all $n \geq 1$. The normalization condition is fulfilled if we set

$$\pi_0 = \left(1 + \sum_{n \geq 1} \frac{p_0 \cdots p_{n-1}}{q_1 \cdots q_n} \right)^{-1}$$

if and only if the series

$$\sum_{n \geq 1} \frac{p_0 \cdots p_{n-1}}{q_1 \cdots q_n}$$

is convergent. Therefore this condition is necessary and sufficient for the existence of an invariant density which is necessarily unique.

We state the following result without proof (most common proofs are based either by analytical methods or by the law of large numbers).

Theorem 5.4 *If i is a recurrent aperiodic state then*

$$\lim_{n \rightarrow \infty} p_{ii}^{(n)} = (\mathbb{E}_i[T_i])^{-1}.$$

Definition 5.5 *A recurrent state i is called fast recurrent (or positive recurrent) if $\mathbb{E}_i[T_i] < +\infty$ and slow recurrent (or null recurrent) if $\mathbb{E}_i[T_i] = +\infty$.*

Note that, if i and j are communicating recurrent and aperiodic states, then they are both fast recurrent or both slow recurrent. It suffices to note that we have inequalities like

$$p_{jj}^{(a+n+b)} \geq p_{ji}^{(a)} p_{ii}^{(n)} p_{ij}^{(b)}$$

where $a, b \geq 1$ are chosen in such a way that $p_{ji}^{(a)} > 0$ and $p_{ij}^{(b)} > 0$ and apply Theorem 5.4.

One could also check that a finite state Markov chain has no slow recurrent states. Indeed, if $\mathcal{C}$ is a class with a slow recurrent state i , by the above argument, one can see that all states in $\mathcal{C}$ are slow recurrent so that the sequence $(p_{jj}^{(n)})_{n \geq 1}$ converges to 0 as n goes to $+\infty$ for all $j \in \mathcal{C}$. Moreover, since it is not possible to exit from $\mathcal{C}$ without leaving it forever (and in this case i would not be recurrent), we have

$$\sum_{j \in \mathcal{C}} p_{kj}^{(n)} = 1$$

for all state $k \in I$ and all n . Taking the limit as n goes to $+\infty$ we get a contradiction.

Theorem 5.6 *Let $(X_n)_{n \geq 0}$ be an irreducible aperiodic Markov chain. The following are equivalent:*

1. *All states are fast recurrent,*
2. *There exists an invariant density $(\pi_i)_{i \in I}$ such that, for all states i, j we have*

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \lim_{n \rightarrow \infty} p_{jj}^{(n)} = \pi_j.$$

In the case where the above conditions 1. and 2. hold $(\pi_i)_{i \in I}$ is the unique invariant density for the Markov chain $(X_n)_{n \geq 0}$.

PROOF. 1. $\Rightarrow$ 2. By Theorem 3.3 (renewal equation) we have

$$p_{ij}^{(n)} = \sum_{k=1}^n f_{ij}^{(k)} p_{jj}^{(n-k)}$$

therefore, by Theorem 5.4 (and dominated convergence), calling π_j the limit of the sequence $(p_{jj}^{(n)})_{n \geq 0}$, one finds

$$\lim_{n \rightarrow \infty} p_{ij}^{(n)} = \sum_{k=1}^{\infty} f_{ij}^{(k)} \pi_j = \pi_j.$$

Moreover, for all finite subset F of I , we have the inequality

$$\sum_{j \in F} \pi_j = \lim_{n \rightarrow \infty} \sum_{j \in F} p_{ij}^{(n)} \leq \lim_{n \rightarrow \infty} \sum_{j \in I} p_{ij}^{(n)} = 1$$

so that

$$\sum_{j \in I} \pi_j \leq 1.$$

We now prove that $(\pi_j)_{j \in I}$ is an invariant density for the Markov chain $(X_n)_{n \geq 0}$. Let F be a finite subset of I . Taking the limit as $n \rightarrow +\infty$ in

$$\sum_{k \in F} p_{ik}^{(n)} p_{kj} \leq p_{ij}^{(n+1)}$$

for all $i, j \in I$, we find the inequalities

$$\sum_{k \in F} \pi_k p_{kj} \leq \pi_j$$

and so, by the arbitrariness of F ,

$$\sum_{k \in I} \pi_k p_{kj} \leq \pi_j. \quad (11)$$

Summing on j we also find

$$\sum_{k \in I} \pi_k = \sum_{j \in I} \sum_{k \in I} \pi_k p_{kj} = \sum_{j \in I} \pi_j$$

and inequalities (11) must be equalities.

Iterating *equalities* (11) we find the identity

$$\pi_j = \sum_{k \in I} \pi_k p_{kj}^{(n)}$$

for all $n \geq 1$. Taking the limit as n goes to infinity, we have

$$\pi_j = \pi_j \sum_{k \in I} \pi_k$$

for all state j so that, since $\pi_j > 0$ because j is *fast* recurrent,

$$\sum_{k \in I} \pi_k = 1.$$

Therefore $(\pi_j)_{j \in I}$ is a probability density which is invariant for $(X_n)_{n \geq 0}$.

2. $\Rightarrow$ 1. This follows immediately from Theorem 5.4.

We finally show that, if 1. and 2. hold, then the invariant density is unique. Note that, if μ is another invariant density, then, for all state j we have

$$\mu_j = \sum_{k \in I} \mu_k p_{kj}.$$

Iterating n -times we find the identity

$$\mu_j = \sum_{k \in I} \mu_k p_{kj}^{(n)}.$$

Taking the limit as n goes to infinity

$$\mu_j = \sum_{k \in I} \mu_k \pi_j = \pi_j$$

for all $j \in I$. ■

6 The strong Markov property

Let $(X_n)_{n \geq 0}$ be a Markov chain on $(\Omega, \mathcal{F}, \mathbb{P})$ with set of states I and transition matrix $(p_{ij})_{i,j \in I}$. For all $n \geq 0$ denote by $\mathcal{F}_n$ the smallest sub- σ -algebra of $\mathcal{F}$ containing events

$$\{X_n = i_n, \dots, X_0 = i_0\}, \quad n \in \mathbb{N}, i_n, \dots, i_0 \in I.$$

Clearly $\mathcal{F}_n$ is composed by events for which one can say whether they occurred or not after observation of random variables $X_0, \dots, X_n$. Roughly speaking, $\mathcal{F}_n$ represents the history of the Markov chain up to time n .

Definition 6.1 *The family $(\mathcal{F}_n)_{n \geq 0}$ of sub- σ -algebras of $\mathcal{F}$ is called **natural filtration** of the Markov chain $(X_n)_{n \geq 0}$.*

We now introduce the notion of *stopping time*.

Definition 6.2 *A random variable*

$$T : \Omega \rightarrow \mathbb{N} \cup \{+\infty\}$$

*is a **stopping time**, with respect to the filtration $(\mathcal{F}_n)_{n \geq 0}$, if the event*

$$\{T \leq n\}$$

belongs to the σ -algebra $\mathcal{F}_n$ for all $n \geq 0$.

Note that, for all state j , the time T_j of the first entrance in j

$$T_j = \{n \geq 1 \mid X_n = j\} \quad (\text{with } \inf \emptyset := +\infty)$$

is a stopping time with respect to the natural filtration $(\mathcal{F}_n)_{n \geq 0}$ because, for all $n \geq 1$,

$$\begin{aligned} \{T_j \leq 0\} &= \emptyset, \\ \{T_j \leq n\} &= \bigcup_{\nu=1}^n \{X_\nu = j\}. \end{aligned}$$

In a similar way the time U_j of the first exit from a state j defined by

$$U_j(\omega) = \begin{cases} \min\{n \geq 1 \mid X_n(\omega) \neq j\} & \text{if } X_n(\omega) \neq j \text{ for some } n \in \mathbb{N}, \\ +\infty & \text{otherwise.} \end{cases}$$

is a stopping time with respect to the natural filtration $(\mathcal{F}_n)_{n \geq 0}$ because, for all $n \geq 1$, we have

$$\begin{aligned} \{U_j \leq 0\} &= \emptyset, \\ \{U_j \leq n\} &= \bigcup_{\nu=1}^n \{X_\nu \neq j\}. \end{aligned}$$

One can easily find the distribution of the random variable U_j which provides a useful insight on the behaviour of a Markov chain.

Theorem 6.3 *If $0 < p_{jj} < 1$ the random variable U_j has geometric distribution with parameter $1 - p_{jj}$. The random variable X_{U_j} defined by*

$$X_{U_j} = \sum_{n \geq 1} X_n(\omega) 1_{\{U_j = n\}}(\omega)$$

if $U_j(\omega) < +\infty$ and in an arbitrary way on the set $\{U_j = +\infty\}$ (which has 0 probability), satisfies

$$\mathbb{P}_j \{ X_{U_j} = k \} = \frac{p_{jk}}{1 - p_{jj}}.$$

for all $k \neq j$.

PROOF. The distribution of U_j is immediately found noting that

$$\mathbb{P}_j \{ U_j > n \} = \mathbb{P}_j \{ X_n = j, \dots, X_1 = j \} = (p_{jj})^n$$

for all $n \geq 1$ so that

$$\mathbb{P}_j \{ U_j = n \} = \mathbb{P}_j \{ U_j > n - 1 \} - \mathbb{P}_j \{ U_j > n \} = (1 - p_{jj}) (p_{jj})^{n-1}.$$

The distribution of X_{U_j} is found in a similar way computing (for $k \neq j$)

$$\begin{aligned} \mathbb{P}_j \{ X_{U_j} = k \} &= \sum_{n=1}^{\infty} \mathbb{P}_j \{ X_n = k, X_{n-1} = j, \dots, X_1 = j \} \\ &= \sum_{n=1}^{\infty} p_{jk} (p_{jj})^n = \frac{p_{jk}}{1 - p_{jj}}. \end{aligned}$$

■

Markov property naturally implies that the distribution of U_j is memoryless and so it must be a geometric distribution.

We now concentrate on the so-called *strong* Markov property which, roughly speaking, is the Markov property when one considers past present and future with respect to a stopping time, not a constant time as in the usual Markov property. As a first step we prove the following

Theorem 6.4 *Let $(X_n)_{n \geq 0}$ be a Markov chain on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with transition matrix $(p_{ij})_{i,j \in I}$ and let T be a stopping time with respect to its natural filtration such that $\mathbb{P} \{ T < \infty \} = 1$. For all $\omega \in \Omega$ denote*

$$Y_n(\omega) = \begin{cases} X_{T(\omega)+n}(\omega) & \text{if } T(\omega) < +\infty, \\ i_\infty & \text{if } T(\omega) = +\infty, \end{cases}$$

where i_∞ is an arbitrary state. The collection of random variables $(Y_n)_{n \geq 0}$ is a Markov chain with the same transition matrix $(p_{ij})_{i,j \in I}$.

PROOF. First note that Y_n is a random variable because, for all $n \geq 0$ and $j \in I$, the set $\{Y_n = j\}$ is an event (i.e. it belongs to $\mathcal{F}$) as it can be written in the form

$$\{X_{T+n} = j\} = \bigcup_{1 \leq \nu < \infty} (\{T = \nu\} \cap \{X_{\nu+n} = j\})$$

if $j \neq i_\infty$ and

$$\{T = +\infty\} \bigcup \left(\bigcup_{1 \leq \nu < \infty} (\{T = \nu\} \cap \{X_{\nu+n} = i_\infty\}) \right)$$

if $j = i_\infty$.

To prove the theorem it suffices to check that, for all $n \geq 0$, and all finite family $(i_k)_{0 \leq k \leq n}$ of states such that the event

$$\bigcap_{0 \leq k \leq n} \{Y_k = i_k\}$$

has strictly positive probability and all state j we have

$$\mathbb{P}(\{Y_{n+1} = j\} \mid \bigcap_{0 \leq k \leq n} \{Y_k = i_k\}) = \mathbb{P}\{Y_{n+1} = j \mid Y_n = i_n\} = p_{i_n j}.$$

As a first step we show that

$$\mathbb{P}\{X_{T+n+1} = j, X_{T+n} = i_n, \dots, X_T = i_0\} = p_{i_n j} \mathbb{P}\{X_{T+n} = i_n, \dots, X_T = i_0\}.$$

To this end note that

$$\begin{aligned} & \mathbb{P}\{X_{T+n+1} = j, X_{T+n} = i_n, \dots, X_T = i_0\} \\ &= \sum_{l=1}^{\infty} \mathbb{P}\{X_{T+n+1} = j, X_{T+n} = i_n, \dots, X_T = i_0, T = l\} \\ &= \sum_{l=0}^{\infty} \mathbb{P}(\{X_{l+n+1} = j\} \mid \{X_{l+n} = i_n, X_{l+n-1} = i_{n-1}, \dots, X_l = i_0, T = l\}) \\ & \quad \cdot \mathbb{P}\{X_{l+n} = i_n, X_{l+n-1} = i_{n-1}, \dots, X_l = i_0, T = l\}. \end{aligned}$$

Now, if $n \geq 1$, the event

$$\{X_{l+n-1} = i_{n-1}, \dots, X_l = i_0, T = l\}$$

belongs to the σ -algebra $\mathcal{F}_{l+n-1}$, and, by the usual Markov property, we find

$$\begin{aligned} & \mathbb{P}\{X_{T+n+1} = j, X_{T+n} = i_n, \dots, X_T = i_0\} \\ &= \sum_{l=0}^{\infty} \mathbb{P}(\{X_{l+n+1} = j\} \mid \{X_{l+n} = i_n\}) \\ & \quad \cdot \mathbb{P}\{X_{l+n} = i_n, X_{l+n-1} = i_{n-1}, \dots, X_l = i_0, T = l\} \\ &= p_{i_n j} \sum_{l=0}^{\infty} \mathbb{P}\{X_{l+n} = i_n, X_{l+n-1} = i_{n-1}, \dots, X_l = i_0, T = l\} \\ &= p_{i_n j} \mathbb{P}\{X_{T+n} = i_n, X_{T+n-1} = i_{n-1}, \dots, X_T = i_0\}. \end{aligned}$$

The equality becomes trivial for $n = 0$.

The identity $\mathbb{P}\{Y_{n+1} = j \mid Y_n = i_n\} = p_{i_n j}$ can be checked in the same way. $\blacksquare$

We shall now prove that the Markov chain $Y = (Y_m)_{m \geq 0}$ is also independent with respect to its past up to a stopping time T with respect to conditional probabilities $\mathbb{P}\{\cdot \mid X_T = i, T < +\infty\}$.

Definition 6.5 Let $(X_n)_{n \geq 0}$ be a Markov chain and T be a stopping time of its natural filtration. We call Markov chain stopped at time T the family $X^{|T}$ of random variables $(X_n^{|T})_{n \geq 0}$ defined by

$$X_n^{|T} = X_{T \wedge n}.$$

Lemma 6.6 Let $(X_n)_{n \geq 0}$ be a Markov chain. For all stopping time T , $m, k \geq 0$, $i, j_1, \dots, j_m \in I$ and all event B belonging to the σ -algebra $\mathcal{F}_k$ we have

$$\begin{aligned} & \mathbb{P}(\{X_{k+m} = j_m, \dots, X_{k+1} = j_1\} \mid \{X_k = i\} \cap \{T = k\} \cap B) \\ &= \mathbb{P}\{X_{k+m} = j_m, \dots, X_{k+1} = j_1 \mid X_k = i\}. \end{aligned}$$

PROOF. Probability measures on $\mathcal{F}_k$

$$\begin{aligned} \mathbb{Q}_1(B) &= \mathbb{P}(\{X_{k+m} = j_m, \dots, X_{k+1} = j_1\} \cap \{X_k = i\} \cap \{T = k\} \cap B) \\ \mathbb{Q}_2(B) &= \mathbb{P}\{X_{k+m} = j_m, \dots, X_{k+1} = j_1 \mid X_k = i\} \cdot \mathbb{P}(\{X_k = i, T = k\} \cap B) \end{aligned}$$

coincide because they coincide on events of the form $\{X_k = i_k, \dots, X_0 = i_0\}$ generating the σ -algebra $\mathcal{F}_k$ by the Markov property. $\blacksquare$

Theorem 6.7 Families of random variables Y and $X^{|T}$ are independent with respect to probability measures $\mathbb{P}\{\cdot \mid X_T = i, T < +\infty\}$.

PROOF. It suffices to check that, for all $n, m > 0$ and all $i = j_0, j_1, \dots, j_m, i_0, \dots, i_n \in I$ the events

$$\{Y_m = j_m, \dots, Y_1 = j_1, Y_0 = j_0\}, \quad \{X_{T \wedge n} = i_n, \dots, X_{T \wedge 0} = i_0\}$$

are independent.

We begin by computing

$$\begin{aligned} & \mathbb{P}\{Y_m = j_m, \dots, Y_0 = j_0, X_{T \wedge n} = i_n, \dots, X_0 = i_0 \mid X_T = i\} \\ &= \sum_{k \geq 0} \mathbb{P}\{X_{k+m} = j_m, \dots, X_k = i, T = k, X_{k \wedge n} = i_n, \dots, X_0 = i_0\} / \mathbb{P}\{X_T = i\} \\ &= \sum_{k \geq 0} \mathbb{P}\{X_{k+m} = j_m, \dots, X_{k+1} = j_1 \mid X_k = i, T = k, X_{k \wedge n} = i_n, \dots, X_0 = i_0\} \\ & \quad \cdot \frac{\mathbb{P}\{X_k = i, T = k, X_{k \wedge n} = i_n, \dots, X_0 = i_0\}}{\mathbb{P}\{X_T = i\}} \end{aligned}$$

By Lemma 6.6, for all $k \geq 0$, we have

$$\begin{aligned} & \mathbb{P} \{ X_{k+m} = j_m, \dots, X_{k+1} = j_1 \mid X_k = i, T = k, X_{k \wedge n} = i_n, \dots, X_0 = i_0 \} \\ &= \mathbb{P} \{ X_{k+m} = j_m, \dots, X_{k+1} = j_1 \mid X_k = i \} \\ &= \mathbb{P} \{ Y_m = j_m, \dots, Y_1 = j_1 \mid X_T = i \} \end{aligned}$$

because $(Y_m)_{m \geq 0}$ is a Markov chain with the same transition matrix of $(X_n)_{n \geq 0}$ and $\{X_T = i\} = \{Y_0 = i\}$.

We find then

$$\begin{aligned} & \mathbb{P} \{ Y_m = j_m, \dots, Y_0 = j_0, X_{T \wedge n} = i_n, \dots, X_0 = i_0 \mid X_T = i \} \\ &= \mathbb{P} \{ Y_m = j_m, \dots, Y_1 = j_1 \mid X_T = i \} \\ & \quad \cdot \sum_{k \geq 0} \frac{\mathbb{P} \{ X_T = i, T = k, X_{T \wedge n} = i_n, \dots, X_0 = i_0 \}}{\mathbb{P} \{ X_T = i \}} \\ &= \mathbb{P} \{ Y_m = j_m, \dots, Y_1 = j_1 \mid X_T = i \} \cdot \mathbb{P} \{ X_{T \wedge n} = i_n, \dots, X_0 = i_0 \mid X_T = i \}. \end{aligned}$$

■

We can now prove that a Markov chain returns to a recurrent state i *infinitely many* times.

Theorem 6.8 *Let N_i the random variable with values $\mathbb{N} \cup \{+\infty\}$*

$$N_i = \sum_{1 \leq n < \infty} 1_{\{X_n = i\}}$$

(number of visits in a state i). A state i is recurrent if and only if $N_i = +\infty$ almost surely

PROOF. Consider the sequence $(T_i^{(n)})_{n \geq 1}$ of stopping time inductively defined by

$$\begin{aligned} T_i^{(1)}(\omega) &= \min \{ n \geq 1 \mid X_n(\omega) = i \}, \\ T_i^{(n+1)}(\omega) &= \min \left\{ k > T_i^{(n)}(\omega) \mid X_k(\omega) = i \right\} \end{aligned}$$

(with $\min \emptyset := +\infty$). If i is transient the conclusion follows from a previous result (recall $\sum_{n \geq 0} p_{ii}^{(n)} < +\infty$ for transient states).

If i is recurrent, then, by definition, the event $\{T_i^{(1)} < +\infty\}$ is almost sure. It follows that, for all $n > 0$, we have

$$\mathbb{P}_i \{ N_i \geq n \} = \mathbb{P}_i \{ T_i^{(n)} < +\infty \} = \mathbb{P}_i \{ T_i^{(n)} < +\infty \mid T_i^{(1)} < +\infty \}.$$

Consider now the Markov chain $(Y_n)_{n \geq 0}$ restarted from $T_i^{(1)}$

$$Y_n(\omega) = \begin{cases} X_{T_i^{(1)}(\omega) + n}(\omega) & \text{se } T_i^{(1)}(\omega) < +\infty, \\ i_\infty & \text{se } T_i^{(1)}(\omega) = +\infty, \end{cases}$$

where i_∞ is an arbitrary state. Since the stopping time $T_i^{(1)}$ is almost surely finite and the random variable Y_0 is almost surely equal to i , from Theorem 6.4 sequences of random variables $(X_n)_{n \geq 0}$ and $(Y_n)_{n \geq 0}$ have the same distributions. Moreover, the n -th entrance in i of $(X_n)_{n \geq 0}$ coincides with the $n-1$ -th entrance in i of $(Y_n)_{n \geq 0}$. Therefore we have

$$\mathbb{P}_i \left\{ T_i^{(n)} < +\infty \mid T_i^{(1)} < +\infty \right\} = \mathbb{P}_i \left\{ T_i^{(n-1)} < +\infty \right\}$$

and, recursively,

$$\mathbb{P} \{ N_i \geq n \} = \dots = \mathbb{P} \left\{ T_i^{(1)} < +\infty \right\} = 1.$$

for all n , i.e. $\mathbb{P} \{ N_i = +\infty \} = 1$. ■